

1. Exploratory Data Analysis (EDA) Summary

- The California Housing dataset was loaded, containing **20,640 samples** and **9 features**, including the target variable MedHouseVal (Median House Value).
- No missing values were found in any of the features, indicating a clean dataset for immediate use.
- A correlation matrix was generated to visualize the relationships between variables:
 - MedInc (Median Income) shows the strongest positive correlation with MedHouseVal (0.69), suggesting it is a significant predictor.
 - AveRooms (Average number of rooms) also has a positive correlation (0.15) with MedHouseVal.
 - Latitude and Longitude show strong negative correlations (-0.14 and -0.05 respectively) with MedHouseVal, indicating geographical influence on housing prices.
 - Other features like HouseAge, AveBedrms, Population, and AveOccup show weaker correlations with the target variable.

2. Model Summary

- **Feature Selection:** The MedHouseVal column was designated as the target variable (y), and all other columns were used as features (X).
- **Train-Test Split:** The data was split into training and testing sets with a 70/30 ratio, respectively, using random_state=42 for reproducibility.
 - Training set size: 14,448 samples
 - Testing set size: 6,192 samples
- **Model Training:** A LinearRegression model from sklearn.linear_model was instantiated and trained on the X_train and y_train datasets.

3. Model Evaluation Metrics

After predicting on the test set (y_pred), the model's performance was evaluated using standard regression metrics:

- **Mean Absolute Error (MAE):** 0.5272
 - On average, the model's predictions deviate by approximately
- **R-squared Score (R²):** 0.5958
 - This indicates that approximately 59.58% of the variance in the target variable (MedHouseVal) can be explained by the model, which is a moderate fit.

Visualizations:

- **Actual vs. Predicted Plot:** Shows a positive linear relationship between actual and predicted values, although there's noticeable scatter, especially at higher values, indicating some underprediction or noise.
- **Residuals Distribution:** The histogram of residuals is centered around zero, but it appears slightly right-skewed, suggesting the model might be underpredicting some higher values. There's also a peak around zero, which is good, but the tails indicate larger errors in some cases.

4. Improvement Ideas

To enhance the model's performance and robustness, consider the following:

1. **Feature Engineering:** Create new features from existing ones. For instance, combine Latitude and Longitude into geographical clusters, or create interaction terms between MedInc and other features.
2. **Polynomial Features:** Linear Regression assumes a linear relationship. Adding polynomial features could capture non-linear relationships in the data.
3. **Regularization:** Techniques like Ridge or Lasso Regression can help prevent overfitting, especially if there are many features or multicollinearity.
4. **Other Models:** Explore more advanced regression models such as Random Forest Regressor, Gradient Boosting Regressor (e.g., XGBoost, LightGBM), or Support Vector Regressor, which often capture complex patterns better than simple linear models.
5. **Outlier Detection and Handling:** Investigate outliers in the data and consider strategies to handle them (e.g., removal, transformation, or using robust models).
6. **Hyperparameter Tuning:** If using models with hyperparameters, perform cross-validation with grid search or random search to find optimal parameter settings.
7. **Error Analysis:** Further investigate the instances where the model makes large errors (from the residuals plot) to understand underlying patterns or data issues.